Creation of a Node – Circular Singly Linked List

```
#include <stdio.h>
#include <stdlib.h>

/* Node structure */
struct Node
{
    int data;
    struct Node *next;
};

/* Function to create a node */
struct Node* createNode(struct Node *head)
{
    struct Node *newnode, *temp;
    int value;

    newnode = (struct Node*)malloc(sizeof(struct Node));

    printf("Enter data: ");
    scanf("%d", &value);

    newnode->data = value;

    /* If list is empty */
    if(head == NULL)
    {
        head = newnode;
        newnode->next = head;    // Points to itself
    }
    else
    {
        temp = head;

        /* Traverse to last node */
        while(temp->next != head)
        {
            temp = temp->next;
        }

        temp->next = newnode;
        newnode->next = head;    // Link back to head
    }
```

```
    return head;
}

/* Function to display the circular linked list */
void display(struct Node *head)
{
    struct Node *temp;

    if(head == NULL)
    {
        printf("List is empty\n");
        return;
    }

    temp = head;

    printf("Circular Linked List: ");

    do
    {
        printf("%d -> ", temp->data);
        temp = temp->next;
    } while(temp != head);

    printf("(back to head)\n");
}

/* Main function */
int main()
{
    struct Node *head = NULL;
    int n, i;

    printf("Enter number of nodes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        head = createNode(head);
    }

    display(head);

    return 0;
}
```

Example Output

```
Enter number of nodes: 3
Enter data: 10
Enter data: 20
Enter data: 30
Circular Linked List: 10 -> 20 -> 30 -> (back to
head)
```

Functions Used

Function	Purpose
createNode()	Creates and adds a node in circular manner
display()	Prints the circular linked list
main()	Controls program execution